

debates exploded about whether India really needs a CBDC, considering that we already have a pretty advanced digital payment system—for both wholesale and retail payments. According to a CLSA report, digital payments in India rose from \$61 billion in 2016 to \$300 billion in 2021. It is expected to go up further to around \$1 trillion by 2026. It is quite evident that people are comfortable with digital payment systems like United Payments Interface (UPI) and Immediate Payment Service (IMPS). So, the question arises as to why we need digital currencies.

In a November 2021 media interview, Garg explained that though more than 95% of payments by value are digital payments, there is about ₹30,000 billion of currency floating in the country, and it is believed that more than 90% of payments by volume are still physical payments. It is this segment of small-value currency transactions that needs CBDC the most.

“CBDC’s advantage is envisioned in terms of creating frictionless, accessible, and affordable payment mechanisms. While we have digital payment systems today, includ-

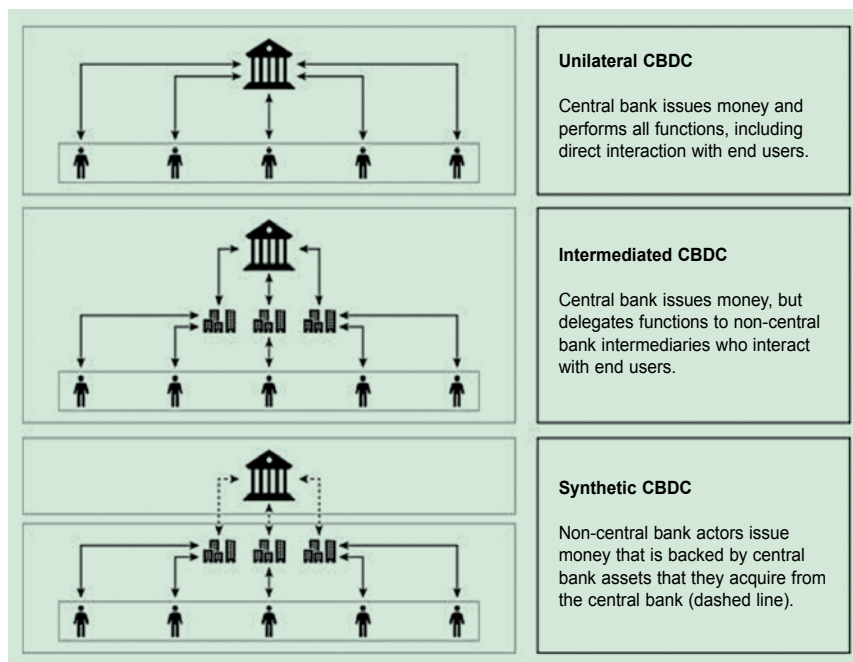


Fig. 3: Three conceptual CBDC operating models (Source: International Monetary Fund)

ing immediate settlements through IMPS, UPI, etc, they are not universal and sometimes carry high fees or transaction costs (particularly cross-border). CBDCs will allow cross-border payments at fractional costs and deliver unprecedented impact,” says Lalwani.

Although the digital rupee is likely to advent in a very simple

form, it will incorporate more features as time goes by.

Understanding retail and wholesale CBDCs

India is likely to introduce both retail and wholesale CBDCs.

Wholesale CBDC is for use by banks and other financial institutions. Wholesale payment systems in India are already digital and pretty well-evolved, so CBDC might not be visibly disruptive but will definitely introduce efficiencies using the blockchain. Wholesale CBDC will enable settlement on net basis. Settlements will be validated using smart contracts. It will be used for interbank, cross-border, capital, and security market settlements. It is rumoured that work on wholesale CBDCs is more advanced in India than retail CBDC.

Retail CBDC is, understandably, more complex because it involves a larger population. It will be used by the general public for conducting day-to-day transactions, usually small-value ones. There are different conceptual operating models (refer Fig. 3), which decide how

What are smart contracts?

A smart contract is a piece of self-executing code, stored on a blockchain, which gets executed when certain predetermined conditions are met. The terms and conditions are directly coded, and exist across a distributed blockchain network. Hence, smart contracts make transactions traceable and trackable but irreversible.

Smart contract can be used to automate the execution of an agreement so that all parties are immediately aware of the outcome without any delay or intermediaries. It can also be used to automate a workflow, triggering actions one after the other as conditions are met. The actions could be like releasing funds, registering properties, or levying fines! The blockchain is updated as soon as the transaction is completed, so it cannot be tampered with.

Smart contracts...

- Are digital and automated, involve no paperwork and reconciliation; hence errors and delays are avoided, and cost is also low
- Do away with third parties; hence no trust issues
- Are secure, because blockchain records are encrypted and hard to hack
- Cannot be tampered with, because each record is linked to the previous and next, and whole chain has to be altered to change a single record

Smart contracts can be used for international trade, retailer-supplier partnerships, medicine or food supply chains, and more. They can help a country to improve enforceability of small contracts.